Forecast uncertainty and simulated storage scheduling

Main reference study and supplementary scenario-count study · v1.1.0

Study scope. A simulated benchmark of forecast-based battery scheduling, with comparisons across risk weights, scenario draws, simulated markets and scenario counts. The main-reference baseline and the supplementary scenario-count study use different sampling layouts, as described below.

Abstract

A battery scheduling benchmark connects seven hourly forecast quantiles to risk-neutral and mean-CVaR decisions. Binary charge/discharge modes enforce exclusive battery operation, while common random numbers compare uncertainty modes using the same underlying draws. The forecast inputs come from Project 1. Across 83 simulated scheduling days, all saved main-reference schedules satisfy full physical diagnostics at tolerance 1e-7. At CVaR weight .55, adaptive scenarios yield mean profit 8.34 EUR/day and empirical 90% CVaR loss 17.90 EUR, compared with raw scenarios at 9.82 and 20.67 EUR. Adaptive scenarios therefore sacrifice mean return for lower tail loss in this particular run; they do not dominate all alternatives. Three scenario-draw seeds and three simulated market seeds expose meaningful sensitivity.

Study design and information timing

Each test day contains 24 scheduling periods. The main-reference baseline input has 1,992 P1 rows, 83 days, seven ordered quantiles, realised simulated prices, seasonal forecasts, and static/adaptive interval endpoints. Each stochastic strategy receives 60 scenarios. The battery has 2 MWh capacity, 1 MW power, charging and discharging efficiency .94, initial/terminal SOC 1 MWh, and throughput cost 2 EUR/MWh. The CVaR confidence level is .90. Forecast fitting and calibration are documented in the P1 report.

Horizon 1 has issue time equal to target time on every test day. No prior-day forecast publication or market gate-closure timestamp is supplied. The experiment is consequently a **simulated 24-hour scheduling benchmark**; it does not establish operational day-ahead availability, implement bid curves or measure realised market revenue.

Strategy, weight .55	Mean profit	5th percentile	CVaR loss	Mean regret
Seasonal naive	-28.28	-80.73	82.64	84.93
Median risk neutral	8.69	-22.75	24.00	47.95
Raw CVaR	9.82	-15.23	20.67	46.82
Static CVaR	8.31	-13.54	18.46	48.33
Adaptive risk neutral	7.63	-21.95	27.39	49.02
Adaptive CVaR	8.34	-14.50	17.90	48.30
Perfect-foresight oracle	56.65	26.24	-24.68	0.00

All amounts are simulated EUR/day except the percentile and CVaR, which are EUR over daily outcomes. Regret is oracle profit minus strategy profit. A negative CVaR loss means that even the worst fixed tail mass remains profitable.

Battery model

For period t, charge c_t and discharge d_t are non-negative MW, SOC e_t is MWh, and mode z_t is binary. With one-hour periods, the model enforces:

0 <= c_t <= P*z_t; 0 <= d_t <= P*(1-z_t); z_t in {0,1}

e_(t+1) = e_t + eta_c*c_t - d_t/eta_d; 0 <= e_t <= E

e_0 = e_T = E*initial_soc_fraction

Each scenario s gives profit pi_s = sum_t[p_(s,t)*(d_t-c_t) - k*(c_t+d_t)]. All scenarios share one schedule. The minimisation objective is negative mean scenario profit plus weight * CVaR_alpha(-pi). SciPy's milp interface calls HiGHS with relative MIP gap 1e-9. Binary modes enforce exclusivity inside the optimization; solver schedules are not clipped, cancelled or otherwise repaired after solving. SciPy documents the integrality and bound interface used here [2,3].

Why exclusivity cannot be assumed

At a single price of -100 EUR/MWh, a lossy simultaneous cycle can be profitable in the continuous model while restoring terminal SOC. The reproducible regression uses E=2, P=1, both efficiencies .94, initial/terminal SOC=1 and cost=2:

Formulation	Charge MW	Discharge MW	End SOC MWh	Profit EUR
Continuous LP relaxation	1.0000	.8836	1.0000	7.8728
Binary MILP	.0000	.0000	1.0000	.0000

The LP relaxation permits simultaneous charging and discharging, so it cannot enforce the operating restriction assumed here. The optional lp_relaxation formulation provides this comparison; the default model is the binary MILP.

Fixed-probability-mass CVaR

Rockafellar-Uryasev CVaR minimises zeta + mean(max(loss_s-zeta,0))/(1-alpha) [1]. The empirical estimator sorts losses from worst to best and averages exactly (1-alpha)*n observations, weighting the boundary observation fractionally. Thus 60 scenario losses use six observations, while 83 evaluation-day losses use 8.3. Ties must not expand that probability mass. Independent LP comparisons, fractional-mass and near-tie tests verify the estimator.

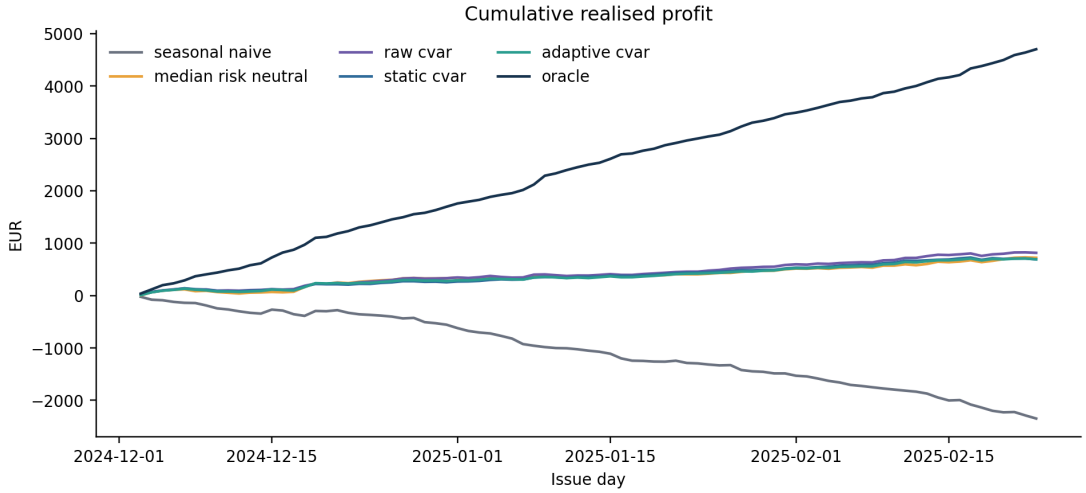
Full diagnostics check finite values, non-negativity, power bounds, all SOC endpoints, every SOC transition and inter-period continuity, initial/terminal SOC and exclusive operation. Counts are reported separately and in aggregate; tolerance is 1e-7.

Scenario construction and comparative results

Scenario uniforms use one shared normal factor and independent hourly factors: latent = .65*common + sqrt(1-.65^2)*idiosyncratic. Distinct latent hours therefore have Gaussian Pearson correlation .4225. This exchangeable dependence structure is a modeling assumption. It does not estimate hour-specific temporal dynamics.

Each hour interpolates the seven forecast quantiles at the latent rank. Values beyond q05/q95 use the boundary quantiles. Static and adaptive curves rescale deviations about q50 by clip(interval_width / max(q90-q10,1e-6), .35, 4). This is a width-based heuristic, not the transformation of conformal coverage into a calibrated joint price distribution. It need not reproduce asymmetric interval endpoints. In the main-reference baseline, the upper cap affects one static and three adaptive rows; no source interval is infinite. Other valid P1 outputs may have infinite intervals when a finite-sample order is unavailable: P2 explicitly counts them and caps their scale at four, discarding interval equivalence.

For each day, the seed is scenario_seed + 17*day_index. Raw, static and adaptive curves use identical uniforms. All frontier weights reuse the same draws as the main adaptive strategy. Thus changing an uncertainty mode or weight does not introduce an unrelated random sample.



Cumulative simulated profits, baseline scenario seed 20260817.

CVaR weight	Raw mean / tail	Static mean / tail	Adaptive mean / tail
.55	9.82 / 20.67	8.31 / 18.46	8.34 / 17.90
.70	9.63 / 20.12	8.19 / 18.37	8.17 / 17.95
.85	9.51 / 19.67	8.64 / 16.86	8.41 / 16.62

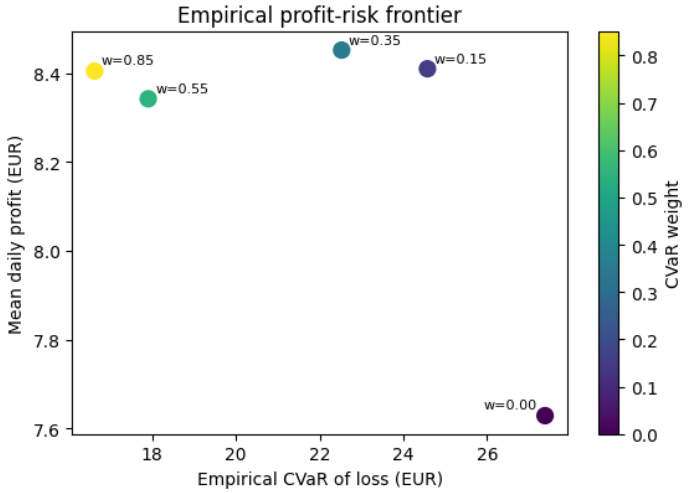
Cells contain mean daily profit / empirical CVaR loss in EUR. Observed out-of-sample profit and tail loss need not vary monotonically with the in-sample penalty. The raw strategy has higher mean profit in all three baseline comparisons; the adaptive strategy has lower tail loss. The oracle has future information and is only a diagnostic upper bound.

Sensitivity to finite scenario draws

Three scenario-draw seeds (20260817, 20260818, 20260819) use the same main-reference baseline P1 forecasts and market outcomes. Each run uses 60 scenarios and 83 full scheduling days. These are Monte Carlo sensitivity checks, not three independent market backtests. Numbers below are averages and ranges over these three seeds; they are not confidence intervals.

Weight	Strategy	Profit avg (min-max)	CVaR avg (min-max)
.55	Raw	9.11 (7.79-9.82)	23.24 (20.67-26.90)
.55	Static	8.25 (7.41-9.04)	19.81 (17.61-23.34)
.55	Adaptive	8.48 (7.48-9.60)	19.59 (17.70-23.18)
.70	Raw	9.10 (7.92-9.74)	22.43 (20.12-24.92)
.70	Static	8.36 (7.50-9.38)	17.70 (15.30-19.45)
.70	Adaptive	8.56 (7.94-9.55)	18.53 (16.49-21.17)
.85	Raw	9.13 (7.98-9.91)	21.57 (19.67-23.19)
.85	Static	8.55 (7.77-9.24)	16.13 (13.87-17.65)
.85	Adaptive	8.61 (8.01-9.42)	17.64 (15.68-20.60)

The adaptive strategy has lower average tail loss than raw across these seeds, but static has lower average tail loss than adaptive at weights .70 and .85. Hence the study supports a conditional risk-return tradeoff, not universal adaptive superiority. The spread across just three draws also motivates larger scenario sets or explicit scenario-count convergence checks before stronger conclusions.



Adaptive risk frontier on the common baseline scenario draws.

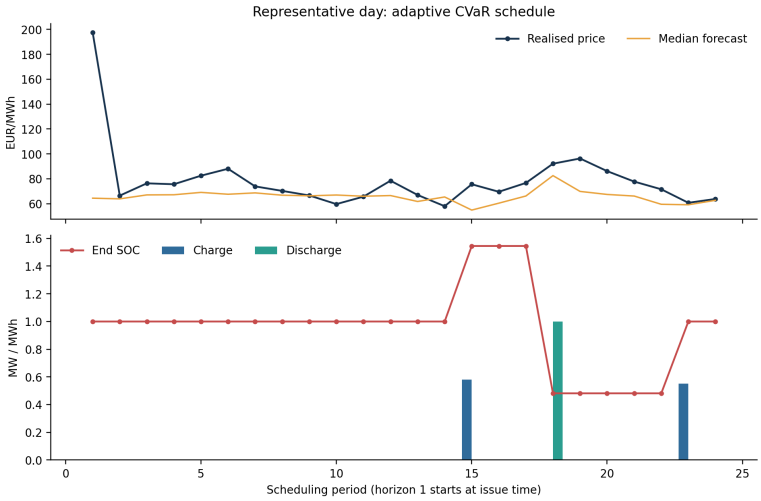
The plotted points describe a discrete empirical evaluation curve, not a certified efficient frontier. Mean profit rises and falls because realized benchmark outcomes differ from the scenario objective.

Sensitivity across three simulated market seeds

Two additional P1 exports regenerate the simulated market and refit the forecast/calibration pipeline with seeds 20260818 and 20260819. They each provide 1,992 rows for the same 83 test-day positions. P2 keeps scenario seed 20260817, CVaR weight .55, 60 scenarios and identical battery settings. These checks vary the simulated data realization; they do not substitute for independent real-market validation.

Market seed	Raw mean / tail	Static mean / tail	Adaptive mean / tail
20260817	9.82 / 20.67	8.31 / 18.46	8.34 / 17.90
20260818	10.22 / 25.89	8.71 / 19.80	8.78 / 20.53
20260819	11.57 / 21.29	10.80 / 18.24	11.04 / 17.11

Cells contain mean daily profit / CVaR loss in EUR. Adaptive has lower observed tail loss than raw on all three simulated market seeds. Static has lower tail loss than adaptive on seed 20260818, while raw has higher mean profit on each seed. Three synthetic realizations do not establish statistical significance or generalization across electricity markets.



Representative exclusive adaptive CVaR schedule in the baseline simulated market.

Supplementary idle and scenario-count study

This study fixes the main-reference baseline simulated market, battery, alpha .90 and weight .55. Only static/adaptive CVaR are compared, for N=60/200, three paired scenario seeds (20260817/18/19) and all 83 days. For each day/seed, one N=200 uniform array is drawn; N=60 takes its exact prefix, and both modes share the same ranks. All 249 prefixes and their hashes are recorded.

The N=60 prefix differs from the main-reference sampling layout. These results are therefore a separate scenario-count study, not a repetition or replacement of the main-reference baseline. All 12 individual results and ranges are retained under corrected_v110/supplementary_20261021/.

N	Strategy	Mean profit avg	CVaR loss avg	Solve seconds
60	Static	8.217827	19.697112	2.833
200	Static	9.712681	14.668371	5.605
60	Adaptive	8.242171	20.727489	2.290
200	Adaptive	9.519176	17.841339	5.014

Profit is simulated EUR/day; CVaR is EUR over daily outcomes. Averages are over three seeds. Timing totals optimization calls for each row across all seeds/days, excludes scenario generation and I/O, and is machine/load dependent. Main-loop wall time was 17.194 seconds.

Seed	Strategy	Profit change	CVaR change	Action change
20260817	Static	+2.491519	-9.562180	1.834897
20260818	Static	+0.462989	-0.874846	1.674168
20260819	Static	+1.530054	-4.649199	1.793132
20260817	Adaptive	+2.523262	-4.816999	1.530517
20260818	Adaptive	+0.202391	-2.525817	1.391540
20260819	Adaptive	+1.105363	-1.315634	1.624484

Changes are N=200 minus N=60. Action change is mean daily L1 charge-plus-discharge change in MWh; 6.361-7.446 hours/day change above 1e-7 MW. All six pairs improve mean profit and CVaR here, but adaptive seed 20260817's fifth-percentile profit worsens from -18.345239 to -20.323186 EUR. Two counts and three seeds do not prove convergence or general superiority.

The zero-transaction alternative holds initial SOC and earns exactly zero profit and CVaR loss on all 83 days. It clarifies that negative-profit trading can underperform feasible inaction. The main-reference seven-strategy outputs are retained separately. All 996 strategy schedules and 83 idle schedules pass full diagnostics: 1,079 schedules, 25,896 periods, zero violations. This bounded synthetic check provides neither calibrated joint scenarios nor real-market validation.

Verification, boundaries and reproducibility

The verification suite has 35 unit tests covering the solver, empirical CVaR, input validation, configured alpha reporting, shared sampling, output protection and supplementary study. The three-weight release verifier compares 18 tables at absolute tolerance 1e-8 and checks 3,486 saved/rerun schedules (83,664 periods). Current local check results and environment details are linked from VERIFICATION.md. Repository CI results apply to the checked commit in its Actions tab.

The scenario-draw and market-seed results comprise 11 unique complete runs, with 6,391 daily schedules and 153,384 periods distributed under corrected_v110/. These files extend the scenario and simulated-market comparisons beyond the three configurations rerun by the release verifier; they are not additional reruns in the current check record.

Battery sensitivity uses only median risk-neutral scheduling on the first 28 days, not adaptive CVaR over all 83 days. Its scope is separate from the full-period risk-weight and scenario studies above.

Practical limits

No observed market or grid data are used. Simulated regimes, tails and dependence may differ from real markets. Marginal intervals do not establish joint path calibration; tail/width clipping also prevents a conformal coverage interpretation for scenarios. The price-taking battery assumes perfect hourly execution and linear throughput cost, omitting network limits, bids, imbalance, reserves, gate closure and detailed degradation. Deployment would require publication-time-safe real inputs and independent-period validation of the whole pipeline.

Reproduction and evidence

Follow the README for fresh local_runs/ destinations and explicit same-type overwrite permission. Source and reference paths are protected. Study scripts default to fresh execution; --reuse-reference validates and labels supplied caches. The baseline input SHA256 is recorded in the input contract and run manifests. PUBLICATION_MANIFEST.json inventories the distributed files. Method sources are credited in RELATED_WORK.md; PUBLICATION.md describes the distributed files and experiment records.

References

[1] Rockafellar, R. T. and Uryasev, S. (2000). Optimization of Conditional Value-at-Risk. Journal of Risk, 2(3), 21-41. Original author-hosted paper: https://sites.math.washington.edu/~rtr/papers/rtr179-CVaR1.pdf

[2] SciPy. scipy.optimize.milp: mixed-integer linear programming, integrality/bounds/constraints and HiGHS interface. Official documentation: https://docs.scipy.org/doc/scipy/reference/generated/scipy.optimize.milp.html (consulted 18 October 2026; local experiments used SciPy 1.17.0).

[3] HiGHS. Official project documentation and solver description: https://highs.dev/ (consulted 18 October 2026).